MECHANICAL AND MECHATRONICS ENGINEERING DEPARTMENT (MME)

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Programs Offered

The Mechanical and Mechatronics Engineering (MME) Department offers two programs – Mechanical Engineering (MECH) and Mechatronics Engineering (MECA). Each program grants three degrees: Bachelor of Science, Bachelor of Engineering, and Master of Science. The details of each program follow.

Vision

To be recognized for the ability, agility, and tenacity of its graduates in confronting the challenges of a changing workplace.

MECHANICAL ENGINEERING PROGRAM

Mission

The mission of the Mechanical Engineering program at RHU is to provide a solid education that enables students to start a professional career and/or continue graduate studies in mechanical engineering. The program seeks to instill in students a sense of and understanding of moral and professional responsibilities, the ability to learn more, and the leadership required in their calling, primarily in Lebanon and the Middle East.

Objectives

In a few years after graduation, Mechanical Engineering program graduates will:

1. Apply their multidisciplinary knowledge in Mechanical engineering to meet employers' expectations.
2. Possess self-learning motivation and qualifications to pursue advanced studies.
3. Demonstrate professional skills and embrace leadership roles in their profession and/or communities.

Program Outcomes

Upon successful completion of the BS/BE program in Mechanical Engineering, graduates will have:

1. Ability to identify, formulate, and solve complex engineering problems by applying principles of engineering, science, and mathematics
2. Ability to apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors
3. Ability to communicate effectively with a range of audiences
4. Ability to recognize ethical and professional responsibilities in engineering situations and make informed judgments, which must consider the impact of engineering solutions in global, economic, environmental, and societal contexts
5. Ability to function effectively on a team whose members together provide leadership, create a collaborative and inclusive environment, establish goals, plan tasks, and meet objectives
6. Ability to develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions
7. Ability to acquire and apply new knowledge as needed, using appropriate learning strategies.

Accreditation

The Bachelor of Engineering program in Mechanical Engineering is accredited by the Engineering Accreditation Commission of ABET (https://www.abet.org) under the commission's General Criteria and Program Criteria for Mechanical and Similarly Named Engineering Programs.

Career Opportunities

Mechanical engineers attain a broad spectrum of skills sought after by a wide range of professions. Industrial sectors, enterprises, and services in which a mechanical engineer can pursue a career include power generation and distribution, building and construction, medicine and pharmacology, aerospace, automotive, chemicals, computers and electronics, renewable energy, entertainment, water resources, sports, environmental institutions, and government.

Mechanical engineering also serves as an excellent foundation for careers in business management and business consulting.

Program Overviews

The Mechanical Engineering Program at Rafik Hariri University puts students' best interests first and foremost. Every little experience the student gains represents a block in the building of a competent, confident, purposeful, problem-solving, competitive, responsible, and conscientious individual. This is accomplished through a curriculum and facilities that meet the highest standards, faculty members committed to students' academic and personal growth, and an environment that inspires learning and drives creativity.

The Bachelor of Engineering (BE) program in Mechanical Engineering comprises 152 credit hours across eight regular semesters and three summer semesters. The Bachelor of Science (BS) program comprises 118 credit hours across six regular semesters and three summer semesters. In addition to completing the credit requirements, a BS or a BE degree is conferred upon a student if the earned Cumulative GPA, Major GPA, and the Summative Learning Project grade are all 70 or above.

The required credit hours span three categories: University requirements (UR), College requirements (CR), and Program requirements (PR). The allocation of the credit hours for the BS and BE programs is shown in the following table:

BS in Mechanical Engineering (118 Credits)						
Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	0	0	21	6	27	23
College Requirements	0	0	24	0	24	20

Program Requirements	49	0	18	0	67	57
Credits	49	0	63	6	118	100

BE in Mechanical Engineering (152 Credits)						
Category	Major		Non-Major		Credits	Percent
	Mandatory	Electives	Mandatory	Electives		
General Education	0	0	21	9	30	20
College Requirements	0	0	27	0	27	18
Program Requirements	59	15	21	0	95	62
Credits	59	15	69	9	152	100

I. University Requirements (General Education)

The list of the University-required courses and their descriptions is presented in the introductory pages of the College of Engineering section in this catalog.

II. College Requirements

The list of college-required courses and their descriptions is presented in the introductory pages of the College of Engineering section in this catalog.

III. Program Requirements

The program requirements for a Bachelor of Engineering degree in Mechanical Engineering

Engineering encompasses 95 credits (67 credits for the BS degree) distributed as follows: 80 credits (67 credits for the BS degree) of mandatory courses and 15 credits for elective courses.

A. Mandatory Requirements

The mandatory courses for the major and non-major Mechanical Engineering programs are listed in the table below.

Course #	Title	Credits	Prerequisites
CIVE 211	Statics	3	
ELEC 320	Electronics	3	ELEC 210
ELEC 320L	Electronics Lab	1	Co-req.: ELEC 320
MECA 340	Dynamic Systems Modeling	3	MATH 314
MECA 341	Measurements and Instrumentation	3	ELEC 320
MECA 341L	Measurements and Instrumentation Lab	1	Co-req.: MECA 340

MECH 400*	Summative Learning (BS) Project	3	ENGL 217
MECA 440	Control Systems Design	3	MECA 340
MECA 440L	Control Systems Design Lab	1	MECA 440
MECH 200	Introduction to Mechanical Engineering	3	
MECH 220	Dynamics	3	CIVE 211
MECH 223	Material Science	3	
MECH 224	Mechanical Engineering Graphics and CAD	3	
MECH 320	Mechanics of Materials	3	CIVE 211
MECH 320L	Mechanics of Materials Lab	1	Co-req.: MECH 320
MECH 321	Mechanisms and Dynamics of Machinery	3	MECH 200 and MECH 220
MECH 330	Thermodynamics I	3	MATH 215
MECH 334	Thermodynamics II	3	MECH 330
MECH 334L	Thermodynamics II Lab	1	Co-req.: MECH 334
MECH 335	Fluid Mechanics	3	MECH 330
MECH 335L	Fluid Mechanics Lab	1	Co-req.: MECH 335
MECH 421	Mechanical Design I	3	MECH 320
MECH 422	Manufacturing processes	3	MECH 320
MECH 423	Mechanical Design II	3	MECH 223, MECH 321, and MECH 421
MECH 431	Heat Transfer	3	MECH 333 or MECH 335 and MATH 317
MECH 431L	Heat Transfer Lab	1	Co-req.: MECH 341
MECH 435	HVAC	3	MECH 431
MECH 499	Co-op Work Experience	3	ENGL 217; Senior Standing
MECH 520	Finite Element Method for Engineers	3	MECH 320 MATH 421 MECH 431
MECH 521	Vibrations and Acoustics	3	MECA 340
MECA 541	CAD/CAM/CAE	3	MECH 224, MECH 421
MECH 595A	BE Summative Learning Project - I	1	ENGL 217 Senior Standing
MECH 595B	BE Summative Learning Project - II	3	MECH 595A
MECH 4xx/5xx	Technical Electives	15	Course requirement

* Required for BS-bound students. BE students take a Technical Elective instead.

B. Technical Electives

As part of the Bachelor of Engineering in Mechanical Engineering program, the student is required to take 15 credit hours of technical electives. These courses allow students to focus on a specific area to develop in-depth knowledge and understanding. The student can also mix and match elective courses from different areas to gain a broader exposure to the various Mechanical Engineering disciplines. The student should, in cooperation with the academic advisor, select a list of electives that best meet their needs and aspirations. The listed technical elective courses and other courses from other engineering programs, in addition to the required program courses, are designed to allow the student to develop in-depth knowledge and understanding in the following areas:

- Mechanical Design and Manufacturing
- Energy and Thermo-fluid Systems

It is highly recommended that the student take elective courses from the list below after completing the mandatory requirements in the related area. Students may select technical electives from other programs, subject to approval of the academic advisor and the corresponding course instructor.

Course #	Title	Credits	Prerequisites
Mechanical Design and Manufacturing			
MECH 445	Automotive Engineering	3	MECH 330 or 333 and MECH 321
MECH 470	Product Design and Development	3	MECH 421
MECH 498	Special Topics in Mechanical Engineering	3	
MECH 522	Mechanism Synthesis: Classical Kinematics to AI-Driven Optimization	3	MECH 321
MECH 523	Mechanics of Composite Materials	3	MECH 320
MECH 524	Elasticity and Plasticity	3	MECH 320
MECH 525	Advanced Dynamics	3	MECH 220
MECH 527	Modern Material Science	3	MECH 223
MECH 570	Digital Manufacturing	3	MECH 422 and MECA 541
MECH 571	Optimization in Engineering Design	3	MATH 421
Energy and Thermo-fluid Systems			
MECH 436	Petroleum Refining Technology for Mechanical Engineers	3	CHEM 211, MECH320, MECH 330
MECH 437	Heat Exchanger Design, Performance, and Inspection	3	MECH 320, MECH 431
MECH 445	Automotive Engineering	3	MECH 330 or 333 and MECH 321
MECH 448	Plumbing Engineering	3	MECH 335 or MECH 333
MECH 451	Fluid Mechanics II	3	MECH 335

MECH 453	Buildings and Energy	3	MECH 330 or MECH 333
MECH 455	Gas Turbines	3	MECH 330 or MECH 333
MECH 461	Thermal Systems Design	3	MECH 431
MECH 468	Internal Combustion Engines	3	MECH 334
MECH 471	Refrigeration	3	MECH 334
MECH 498	Special Topics in Mechanical Engineering	3	
MECH 530	Computational Fluid Dynamics	3	MECH 431
MECH 541	Renewable Energy	3	MECH 330 or MECH 333

C. Summative Learning Project

Students must complete a 3-credit-hour course for the BS (4 credits for the BE; taken 1 credit in the first regular semester and 3 credits in the following regular semester) in which they work, preferably in groups, on a problem of concern to industry or the community at large, or to develop a promising idea. The SLP could be an extension of the projects students worked on in the Engineering for the Community course, bringing them to a more useful outcome.

D. Co-op Experience

Students must complete 8 weeks of practical training in a field related to their interests. The Co-op experience is usually fulfilled during the summer semester of the third year in the program. The Co-op office matches the Co-op assignment with the student's field of study and the employer's interest. A faculty member follows up on a student's progress throughout the Co-op period by conducting field visits and ensuring that the student's performance is aligned with their aspirations and the employer's needs. Students are required to submit a formal report and a poster and to give a formal presentation about their Co-op experience. While ENGL 217 and Senior Standing are the essential prerequisites for co-op training, the program may stipulate additional conditions.

Study Plan

The following study plan summarizes the courses and credit distribution for the Bachelor of Science (BS) and Bachelor of Engineering (BE) in Mechanical Engineering. The following study plan serves as a roadmap for smooth progression toward graduation.

Course #	Title	Credits	Prerequisites
Year 1, Fall Semester (15 Credits)			
CCEE 214	Introduction to Programming	3	
CIVE 211	Statics	3	
ENGL 210	English Composition and Rhetoric	3	Placement
MATH 215	Calculus III	3	
PHYS 210	Fundamentals of Physics and Lab	3	

Year 1, Spring Semester (15 Credits)			
ELEC 210	Electric Circuits	3	PHYS 191 or Equivalent
MATH 216	Calculus IV	3	MATH 215
MECH 200	Introduction to Mechanical Engineering	3	
MECH 220	Dynamics	3	CIVE 211
MECH 223	Material Science	3	
Year 1, Summer Semester (9 Credits)			
CHEM 211	Environmental Chemistry and Lab	3	
MECH 224	Mechanical Engineering Graphics and CAD	3	
MATH 311	Linear Algebra with Applications	3	
Year 2, Fall Semester (17 Credits)			
ELEC 320	Electronics	3	ELEC 210
ELEC 320L	Electronic Lab	1	Co-req: ELEC 320
MATH 314	Ordinary Differential Equations	3	MATH 216
MECH 330	Thermodynamics I	3	MATH 215
MECH 320	Mechanics of Materials	3	CIVE 211
MECH 320L	Mechanics of Materials Lab	1	Co-req.: MECH 320
MECH 321	Mechanisms and Dynamics of Machinery	3	MECH 200 and MECH 220
Year 2, Spring Semester (17 Credits)			
ENGL 217	Professional English	3	ENGL 210
MATH 351	Probability and Statistics	3	MATH 215
MECA 340	Dynamic Systems Modeling	3	MATH 314
MECH 334	Thermodynamics II	3	MECH 330
MECH 334L	Thermodynamics II Lab	1	Co-req: MECH 334
MECH 335	Fluid Mechanics	3	MECH 330
MECH 335L	Fluid Mechanics Lab	1	Co-req.: MECH 335
Year 2, Summer Semester (9 Credits)			
ENGR 300	Engineering Economics and Management	3	
MATH 317	Partial Differential Equations	3	MATH 314
	Social Sciences/Cultural Elective I	3	
Year 3, Fall Semester (16 Credits)			
MATH 421	Numerical Analysis	3	MATH 311
MECA 440	Control Systems Design	3	MECA 340
MECA 541	CAD/CAM/CAE	3	MECH 224, MECH 421
MECH 421	Mechanical Design I	3	MECH 320
MECH 431	Heat Transfer	3	MECH 333 or MECH 335 and MATH 317